

CHATHAM, MA, USA

WMO: 725069

Lat: 41.688N

Lon: 69.993W

Elev: 21

StdP: 101.08

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 94624

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-8.7	-19.5	0.7	-8.9	-17.1	0.8	-6.9	11.4	3.0	10.0	2.9	3.7	300	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	28.6	23.0	27.2	22.2	25.9	21.5	24.2	26.8	23.5	25.7	22.8	24.8	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.1	25.8	22.7	17.4	25.0	22.2	16.9	24.4	72.9	27.0	69.9	25.6	67.2	24.9	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	7.9	7.1		-13.6	31.4	3.3	2.1	-16.0	33.0	-17.9	34.2	-19.7	35.4	-22.1	36.9
				-14.6	25.4	3.0	1.0	-16.8	26.1	-18.6	26.7	-20.3	27.3	-22.5	28.0
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	0.0	0.2	3.2	7.8	12.8	17.8	21.8	21.5	18.1	12.7	7.7	3.0		
	DBStd	8.51	4.84	4.24	3.75	3.14	3.18	3.19	2.62	2.53	2.95	3.74	4.08	4.27		
	HDD10.0	1212	311	276	213	79	9	0	0	0	0	16	90	218		
	HDD18.3	3101	568	509	471	316	173	47	4	4	39	178	318	474		
	CDD10.0	1438	0	0	1	14	97	235	366	357	244	100	22	2		
	CDD18.3	284	0	0	0	0	2	32	111	103	33	3	0	0		
	CDH23.3	1109	0	0	0	3	11	103	499	425	64	3	0	0		
	CDH26.7	174	0	0	0	0	3	17	92	59	4	0	0	0		
(14) Wind	WSAvg	3.3	3.5	3.6	3.9	3.8	3.4	3.2	3.0	2.8	3.0	3.2	3.4	3.5		
(15) Precipitation	PrecAvg	989	86	76	94	92	73	74	57	70	84	83	95	98		
	PrecMax	1266	153	172	223	182	171	264	149	217	191	196	202	177		
	PrecMin	577	19	32	10	27	15	7	11	0	23	14	20	13		
	PrecStd	193	36	34	53	37	42	52	28	47	47	46	49	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	11.0	14.0	19.6	24.6	28.8	31.1	30.1	27.2	23.0	17.5	14.1		
		MCWB	11.2	9.4	9.6	13.0	18.0	22.0	23.6	23.8	22.3	18.9	14.5	12.4		
	2%	DB	9.9	9.1	11.4	15.6	21.2	26.0	29.0	28.1	25.1	20.8	16.2	12.4		
		MCWB	8.8	7.9	8.5	11.3	16.4	20.2	23.1	23.1	20.9	18.4	14.5	11.5		
	5%	DB	8.4	7.7	9.8	13.8	19.1	24.0	27.4	27.0	23.8	19.2	14.8	11.1		
		MCWB	7.0	6.2	7.7	10.2	15.0	19.2	22.2	22.5	20.3	17.1	13.3	10.3		
	10%	DB	6.9	6.3	8.4	12.4	17.6	22.6	26.1	25.8	22.6	17.9	13.7	9.8		
		MCWB	5.5	4.6	6.5	9.7	14.2	18.7	21.7	21.7	19.8	16.1	12.0	8.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	10.3	11.7	15.2	19.2	23.0	25.4	25.4	23.7	20.4	16.1	13.3		
		MCDB	11.6	10.6	12.5	17.6	23.1	27.2	29.7	28.7	26.1	21.9	16.7	13.6		
	2%	WB	9.1	8.3	9.5	12.8	17.6	21.7	24.3	24.4	22.4	19.0	15.0	11.8		
		MCDB	9.6	8.8	10.6	14.6	20.0	24.7	27.2	26.8	24.0	20.0	15.9	12.2		
	5%	WB	7.5	6.8	8.1	11.6	16.3	20.7	23.5	23.7	21.5	17.8	13.6	10.4		
		MCDB	8.2	7.6	9.1	12.7	18.2	22.9	25.9	25.7	22.9	18.8	14.7	10.9		
	10%	WB	5.7	5.1	6.9	10.4	15.0	19.7	22.8	22.9	20.7	16.7	12.4	8.8		
		MCDB	6.7	6.2	8.1	11.8	17.0	21.6	24.9	24.8	22.1	17.8	13.3	9.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.1	6.5	6.7	6.9	7.0	7.1	7.1	6.9	6.9	6.6	6.7		
		MCDBR	7.6	7.8	8.0	8.6	9.3	8.9	8.6	7.9	7.6	7.3	6.6	7.5		
		MCWBR	7.6	7.3	6.3	5.5	5.5	5.0	4.5	4.5	5.0	5.7	5.9	7.8		
		MCDBR	7.5	7.7	7.2	7.3	8.5	7.7	7.3	6.9	6.4	6.7	6.1	7.3		
	5% WB	MCWBR	7.9	7.9	6.6	5.8	5.5	4.8	4.3	4.4	4.9	6.0	6.2	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.320	0.338	0.375	0.408	0.456	0.481	0.440	0.374	0.350	0.323	0.310		
	taud		2.511	2.496	2.482	2.396	2.349	2.245	2.186	2.297	2.457	2.521	2.561	2.559		
	Ebn at Noon		839	891	912	895	869	825	799	823	861	845	825	817		
	Edh at Noon		73	86	97	114	123	137	143	125	99	83	69	65		
(44) All-Sky Solar Radiation	RadAvg		1.67	2.58	3.66	4.78	5.43	5.82	6.00	5.41	4.39	2.98	1.94	1.43		
	RadStd		0.13	0.23	0.29	0.51	0.53	0.52	0.33	0.28	0.32	0.25	0.15	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	+0.54	+0.51	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+97	+47

Nomenclature: See separate page